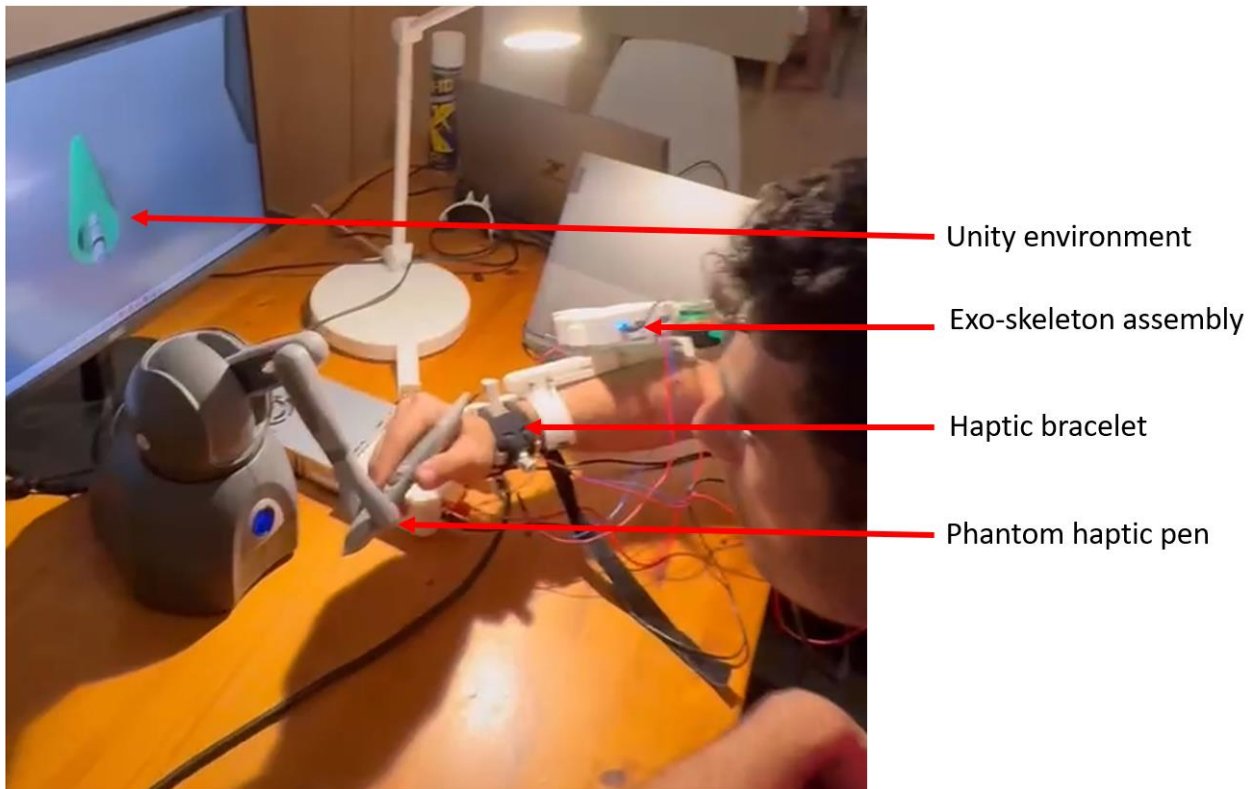


# Haptic Feedback Prototype for Cochlear Implant Electrode Insertion Training

## Repository purpose

This repository documents the design, construction, and testing of a haptic feedback prototype for cochlear implant electrode insertion training. The system combines a Unity-based Phantom Touch simulation, an actuated arm exoskeleton, and four Titan Haptics Drake vibrotactile motors. The goal is to guide the user through a controlled simulated insertion motion while providing real-time feedback about insertion position and insertion speed.

The prototype is not intended for clinical use. It is an educational and experimental proof of concept for exploring how haptic guidance and vibrotactile feedback can support controlled manual motion during a simulated cochlear implant electrode insertion task.



**Figure 1:** Full prototype overview showing the Phantom Unity setup, exoskeleton and haptic motors.

# 1. Introduction

Cochlear implants are surgically implanted hearing prostheses used for people with severe to profound hearing loss when conventional acoustic amplification is insufficient. The implant bypasses damaged hair cells by electrically stimulating the auditory nerve through an electrode array inserted into the cochlea [1], [2]. Because the electrode is inserted into a delicate spiral-shaped inner-ear structure, the insertion process must be performed carefully. Excessive insertion force, unstable insertion speed, poor trajectory, or excessive insertion depth can increase the risk of intracochlear trauma and may negatively affect residual hearing preservation [3]–[6].

In modern cochlear implant surgery, preserving residual hearing is an important objective. Several studies report that insertion mechanics influence forces and pressure variations inside the cochlea. Kontorinis et al. showed that higher electrode insertion speeds can increase insertion forces in an artificial scala tympani model [5]. More recent work also reports that insertion speed and operator experience influence intracochlear pressure variation during simulated cochlear implantation [6]. These findings make insertion control a relevant target for training and feedback systems.

A key challenge is that cochlear implant insertion is a fine manual task. The surgeon must control the electrode trajectory, speed, and final depth while operating at a very small anatomical scale. Although experienced surgeons develop tactile skill through practice, novice users may not immediately understand how their hand motion relates to safe insertion behaviour. Existing solutions include surgical training, physical cochlear models, virtual reality simulators, and robotic insertion systems [3], [7]–[10]. However, many of these systems are expensive, complex, or focused mainly on visual feedback and force measurement rather than simple real-time tactile feedback that can guide the user during the movement.

Haptic technology provides a unique advantage because it can communicate information through touch without requiring the user to constantly look away from the task. Kinesthetic haptic guidance can physically guide or resist the user's motion, while vibrotactile feedback can communicate warnings, position zones, or speed levels through small contact-based motors. Prior research in surgical robotics and motor learning suggests that haptic and vibrotactile feedback can improve motion awareness and training support when designed carefully [11]–[13].

This project addresses the gap between complex surgical simulation systems and simple real-time motion guidance. The proposed prototype combines three feedback layers:

1. **Unity and Phantom Touch environment:** provides the simulated cochlear implant insertion task and movement data from the Phantom pen.
2. **Actuated arm exoskeleton:** guides the user through the insertion motion using an ODrive-controlled motor.
3. **Haptic motor feedback system:** uses four Titan Haptics Drake motors to provide tactile feedback about insertion position and speed.

The central idea is that the user should not only see the simulated insertion environment, but also feel how the insertion should progress. The exoskeleton guides the movement path, while the haptic motors inform the user about position and warn them when their insertion speed is too high.

## 2. Supplies / Bill of Materials

The table below lists the main components required to reproduce the prototype.

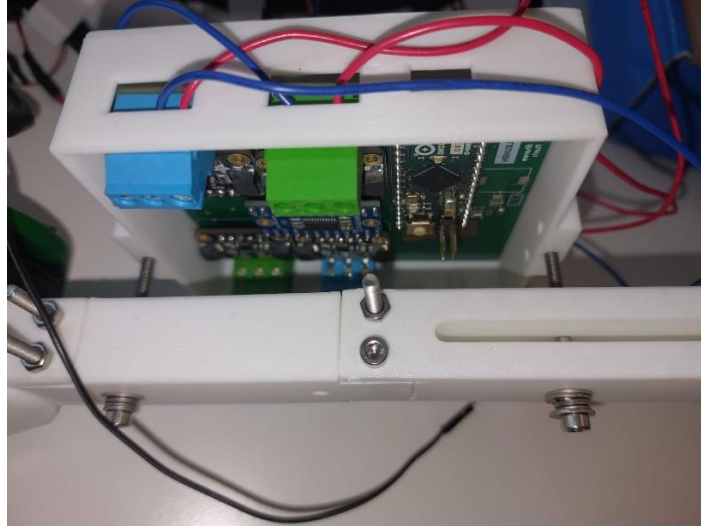
| Component                | Quantity | Example part / model               | Purpose in prototype                                                                 |
|--------------------------|----------|------------------------------------|--------------------------------------------------------------------------------------|
| Phantom haptic pen       | 1        | Phantom haptic pen                 | Provides the Unity-based simulated insertion environment and pen movement data       |
| Arm exoskeleton          | 1        | Custom mechanical structure        | Physically guides the user's arm or hand through the insertion motion                |
| ODrive controller        | 1        | ODrive S1                          | Controls the exoskeleton motor                                                       |
| Exoskeleton motor        | 1        | Motor model D5312S                 | Actuates the exoskeleton and pushes/guides the user through the insertion trajectory |
| External DC power supply | 1        | Voltage/current rating to be added | Powers the ODrive and exoskeleton motor                                              |
| Arduino Micro            | 1        | Atmega32U4, 5 V, 16 MHz            | Main microcontroller for receiving values and controlling the haptic motors          |
| Haptic motor driver      | 4        | DRV2605L or equivalent             | Drives each Drake haptic motor using I2C commands                                    |
| I2C multiplexer          | 1        | TCA9548A 1-to-8 I2C multiplexer    | Allows multiple same-address haptic drivers to be controlled                         |

| Component                   | Quantity | Example part / model                                         | Purpose in prototype                                                  |
|-----------------------------|----------|--------------------------------------------------------------|-----------------------------------------------------------------------|
| Vibrotactile motors         | 4        | Titan Haptics Drake motors                                   | independently<br>Provide tactile feedback for position and speed cues |
| Custom PCB                  | 1        | Soldered printed circuit board                               | Permanent electrical assembly mounted on the exoskeleton              |
| Motor wiring and connectors | 4 sets   | Soldered motor attachments/connectors                        | Connect each Drake motor to the PCB                                   |
| USB cable for Arduino       | 1        | micro-USB                                                    | Powers the Arduino and enables serial communication                   |
| Software tools              | -        | Unity, Arduino IDE, ODrive tools, Python/Excel data handling | Simulation, control, communication, and testing                       |

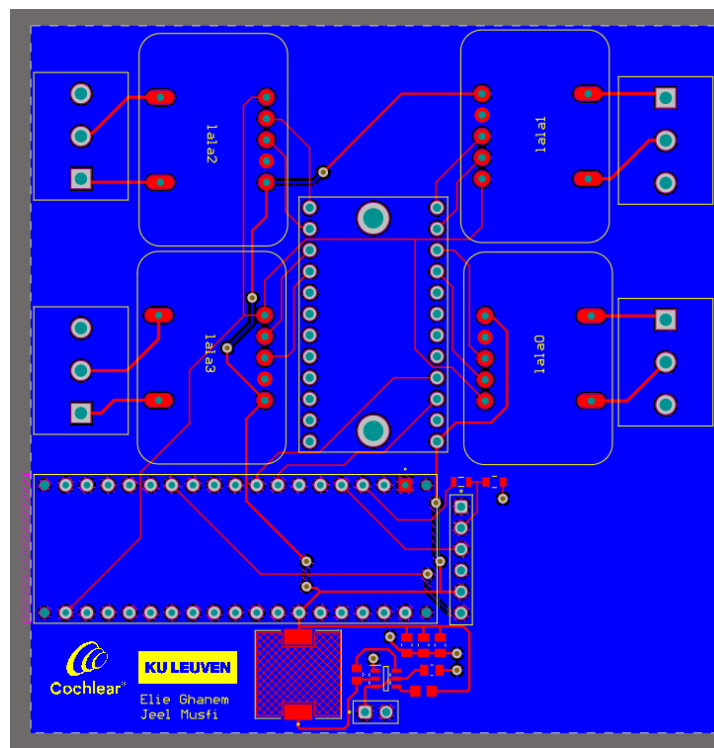
## Hardware notes

The final prototype uses a soldered PCB rather than a breadboard. This was important because the electronics are mounted to the exoskeleton, where movement and vibration can cause loose breadboard connections to fail. The haptic drivers, multiplexer, and related electronics are soldered directly onto the PCB. The four Drake motors are wired to soldered attachments on the PCB so that the motors can still be routed and positioned mechanically.

The Arduino Micro is powered through micro-USB. The ODrive and exoskeleton motor are powered by an external DC power supply. The ODrive is also connected to the laptop through USB-C for configuration and monitoring. The Arduino communicates with the haptic motor electronics through I2C, while the haptic drivers are selected through the TCA9548A multiplexer.



**Figure 2:** Close-up photo of the PCB mounted on the exoskeleton.



**Figure 3:** Wiring diagram of the PCB

### 3. Methods

#### Step 1: Medical challenge definition

The selected medical challenge is controlled cochlear implant electrode insertion. The electrode array must be inserted into the cochlea with a controlled trajectory, speed, and depth. The difficulty comes from the small scale of the cochlea and the importance of minimizing mechanical trauma. The design target of this project was therefore not to perform a real clinical insertion, but to create a training-oriented prototype that helps a user understand and reproduce controlled insertion behaviour through haptic guidance and vibrotactile feedback.

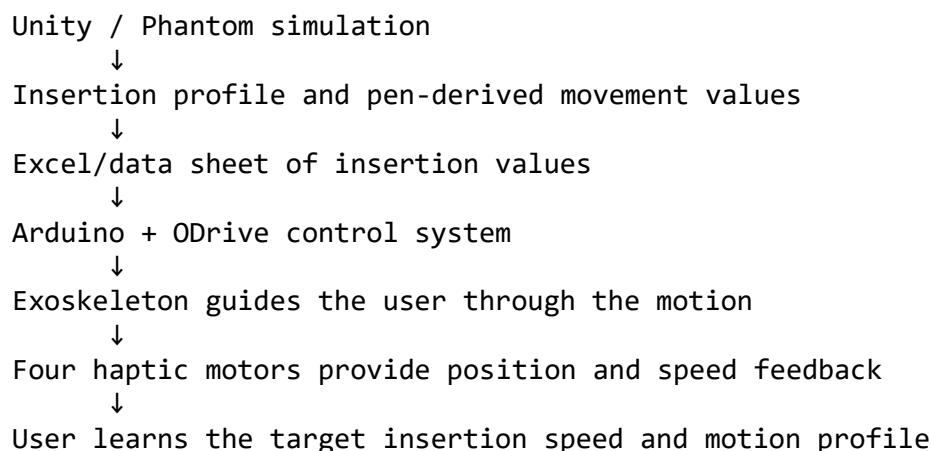
The main variables selected for feedback and guidance were:

- **Insertion depth:** where the user is along the simulated insertion path.
- **Insertion speed:** how fast the user is moving during insertion.
- **Guided motion:** the exoskeleton physically assists the user by guiding the movement along the intended insertion trajectory.

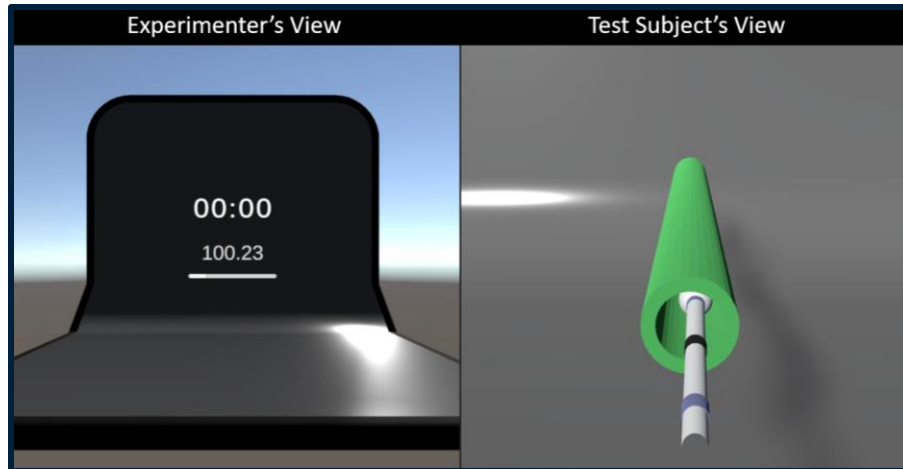
These variables were chosen because the literature identifies insertion speed and insertion mechanics as important factors related to insertion forces, intracochlear pressure changes, and atraumatic insertion technique [3]–[6].

#### Step 2: Conceptual system design

The system was designed as a training loop that combines a simulated insertion task with physical guidance and tactile feedback:



The Phantom Touch is mainly used with its Unity environment. It provides the simulated task context and allows linear movement data from the pen to be retrieved. The final prototype does not rely heavily on the Phantom for physical guidance. Instead, the insertion profile is represented as a set of values, for example in an Excel sheet, which is sent to the exoskeleton and haptic feedback system. This allows the prototype to reproduce a planned insertion profile and guide the user through it repeatedly.



**Figure 4:** Unity Phantom insertion environment.

### Step 3: Phantom and Unity integration

The Phantom Touch is used in a Unity environment to represent the cochlear implant insertion task. The relevant output from this environment is the linear movement of the pen, which is used to define or extract insertion speed. The target maximum insertion speed used in this project is **1 mm/s**.

For repeatable training, an Excel sheet of insertion values is used as the reference input to the exoskeleton and wearable haptic system. This makes the same insertion trajectory repeatable across trials. The linear speed from the pen is used to define the desired insertion behaviour, while the haptic motors communicate whether the user is moving within or beyond the allowed speed range.

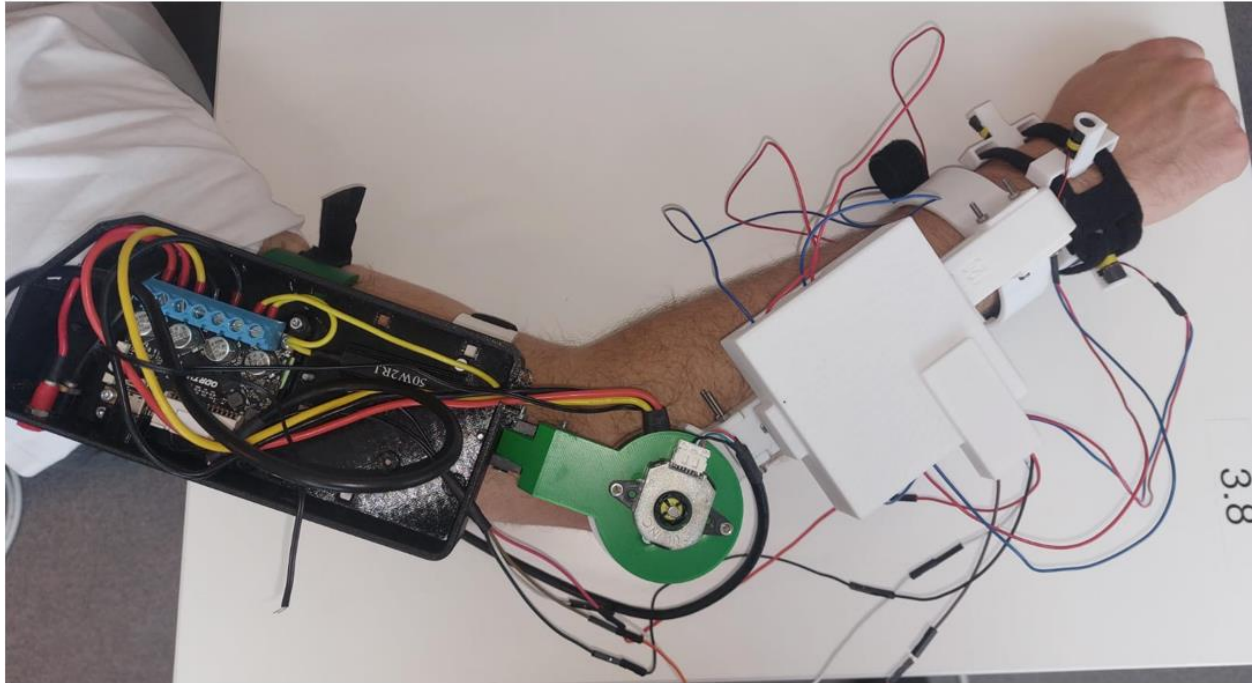
### Step 4: Exoskeleton design and function

The exoskeleton is the main physical guidance element of the prototype. Its role is to guide the user's movement and push them through the desired insertion path. This makes the system more active than a passive stylus holder or simple wrist support. The user still interacts with the system, but the exoskeleton helps shape the motion so that the insertion follows the intended profile.

The design requirements were:

- guide the user through the desired insertion movement
- keep the movement controlled and repeatable
- avoid unsafe or sudden actuation
- allow the user to feel the guided motion
- leave enough space for the haptic motor feedback system and PCB
- be mechanically compatible with the Phantom/Unity training setup
- allow testing and retuning of the control gains

The exoskeleton is actuated by a motor controlled using an ODrive motor controller. The motor actuates the elbow joint as can be seen in figure 5. The ODrive is powered by an external DC power supply and connected to the laptop through USB-C for configuration and monitoring. The control values were tuned experimentally using trial and error until the movement became stable and sufficiently responsive.



**Figure 5:** Exo-skeleton assembly on arm of the user.

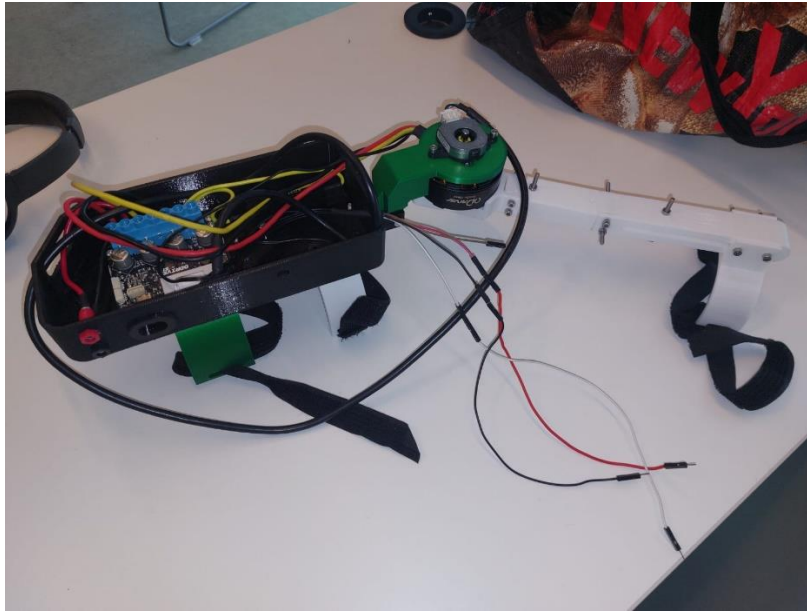
The final experimentally selected control gains were:

| Parameter                | Final value | Reasoning                                                                                                                        |
|--------------------------|-------------|----------------------------------------------------------------------------------------------------------------------------------|
| Position gain            | 42.5        | High enough to guide the motion clearly without excessive oscillation                                                            |
| Velocity gain            | 0.315       | Added damping and reduced unstable movement during tracking                                                                      |
| Velocity integrator gain | 0           | Integral action was removed because it was not needed for the guided motion and could increase unwanted vibration or instability |

The tuning process started from low gains and increased the position gain until the system could guide the movement effectively. The velocity gain was then adjusted to add damping and



reduce vibration. The velocity integrator was left at zero because the goal was not perfect long-term position correction but stable, safe guidance through a repeated insertion profile.



**Figure 6:** Exoskeleton mechanical structure.



**Figure 7:** Motor connection to the exoskeleton.

## Step 5: Haptic feedback motor layout

The haptic feedback system uses four Titan Haptics Drake motors. The motors provide two types of information:

1. **Position feedback:** which part of the insertion cycle the user is currently in.
2. **Speed feedback:** whether the insertion speed is safe, approaching the limit, or too fast.

The position feedback is based on a cyclic angular mapping. The current position is wrapped into a 0°-360° range. If the value exceeds 360°, the same motor pattern repeats using modulo 360°. Four motors are used, but the transition regions activate two motors at the same time. This gives a smoother feeling of progression between motor locations.

| Position range | Active motor feedback      |
|----------------|----------------------------|
| 0°–45°         | Motor 1 only               |
| 45°–90°        | Motor 1 + Motor 2          |
| 90°–135°       | Motor 2 only               |
| 135°–180°      | Motor 2 + Motor 3          |
| 180°–225°      | Motor 3 only               |
| 225°–270°      | Motor 3 + Motor 4          |
| 270°–315°      | Motor 4 only               |
| 315°–360°      | Motor 4 + Motor 1          |
| >360°          | Mapping repeats cyclically |

This mapping was selected because it allows four physical motors to represent more than four position states. Single-motor activation gives a clear location cue, while dual-motor activation indicates that the user is between two position regions.

## Step 6: Speed feedback mapping

The maximum desired insertion speed is **1 mm/s**. The haptic motors communicate speed using pulse frequency and continuous vibration:

| Speed range   | Feedback behaviour   | Meaning                            |
|---------------|----------------------|------------------------------------|
| 0–0.25 mm/s   | Slow pulses          | Very slow / controlled movement    |
| 0.25–0.5 mm/s | More frequent pulses | Safe movement, slightly faster     |
| 0.5–0.75 mm/s | Frequent pulses      | Approaching the upper safe range   |
| 0.75–1.0 mm/s | Very frequent pulses | Close to the maximum allowed speed |
| >1.0 mm/s     | Continuous vibration | Too fast, warning feedback         |

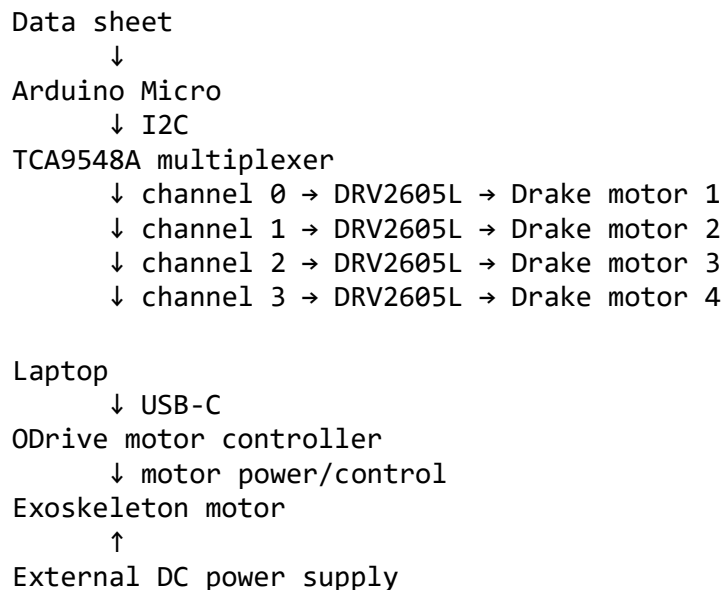
The feedback logic was designed so that the user can feel the speed increase without looking at a display. Below the maximum speed, the feedback remains pulse-based. The closer the user gets to 1 mm/s, the more pulses they feel. If the speed exceeds 1 mm/s, the motor changes to continuous vibration to clearly warn that the insertion is too fast.

The intensity is also adjusted depending on whether one or two motors are active. Single-motor feedback can use stronger intensity because the vibration is localized. When two motors are active, the intensity is reduced so that the combined sensation is not too strong or confusing.

## Step 7: Electronics integration

The electronics were built around an Arduino Micro and a soldered PCB. The Arduino receives insertion profile values and controls the haptic motors through the TCA9548A multiplexer and DRV2605L haptic drivers. The ODrive controls the exoskeleton motor.

The electronics architecture is:



The TCA9548A multiplexer is needed because the DRV2605L haptic drivers use the same I2C address. The multiplexer allows the Arduino to select one driver channel or two driver channels depending on the active position sector. This enables both single-motor and dual-motor feedback.

The soldered PCB improves robustness compared with a breadboard. Since the electronics are attached to a moving exoskeleton, a breadboard would be more likely to suffer from loose wires, poor contact, or vibration-related disconnection.

## Step 8: Firmware and control logic

The firmware combines two main functions: exoskeleton actuation and haptic motor control.

For the exoskeleton, the incoming depth or trajectory value is converted into an ODrive target position. The ODrive is configured for position control and the exoskeleton moves toward the target. A timeout is included so that if no new data is received for a defined period, the exoskeleton can be released for safety.

For the haptic motors, the firmware wraps the current position into a  $0^{\circ}$ – $360^{\circ}$  range and selects the appropriate motor or motor pair. Then it checks the speed value and applies either pulse feedback or continuous vibration. At the end of the insertion range, a completion pattern can be played by cycling through the four motors.

## 4. Testing and results

### Haptic motor feedback testing

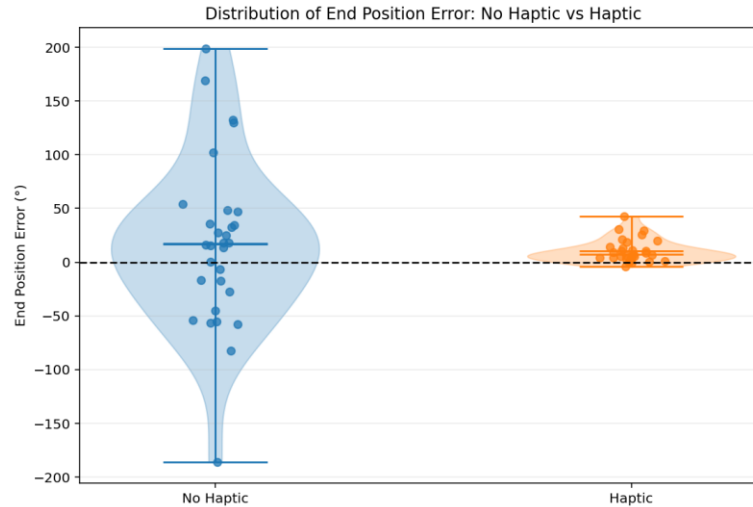
The haptic feedback system was tested separately from the complete exoskeleton assembly. The purpose of this test was to confirm that the four Drake motors could communicate position and speed information clearly.

The position feedback test confirmed that the motors could be activated individually and in pairs according to the angular mapping. This made it possible to represent eight position sectors using only four physical motors.

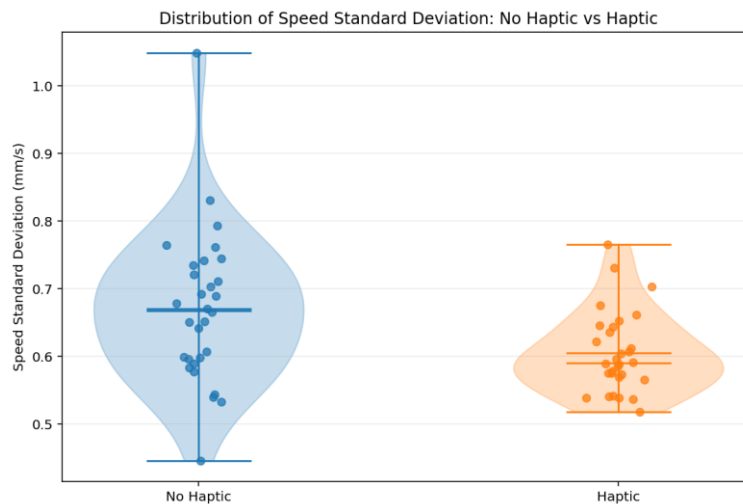
The speed feedback test confirmed that the vibration pattern changed depending on speed. Below 1 mm/s, the feedback was pulse-based. As the speed approached 1 mm/s, the number of pulses increased and the pauses between pulses decreased. Above 1 mm/s, the selected motor changed to continuous vibration, which acted as a clear warning that the insertion speed was too high.



**Figure 8:** Haptic device mount for testing only



**Figure 9:** End-position results



**Figure 10:** Speed standard deviation results

The end-position error results show that the haptic feedback condition produced a narrower distribution of final-position errors compared with the no-haptic condition. This suggests that the tactile cues helped the user identify the intended end region more consistently. The no-haptic condition showed a wider spread and larger deviations, meaning that relying only on visual or internal estimation made the final position less repeatable.

The speed standard deviation results also show a slightly narrower distribution with haptic feedback. This suggests that the pulse-based speed feedback helped the user maintain a more consistent insertion speed. Since the goal of the system is not only to reach the final position but also to teach controlled motion, this supports the use of vibrotactile feedback as a training aid. However, these results should be interpreted as preliminary because the test was performed on the haptic feedback system only and not yet on the complete exoskeleton-integrated assembly.

## Exoskeleton testing

Full combined testing of the complete assembly should still be done. This should include the exoskeleton, PCB, haptic motors, ODrive, motor, power supply, and Unity/Phantom data source running together.

## 5. Troubleshooting and iterative improvements

### I2C address conflict

All DRV2605L haptic drivers use the same default I2C address. This prevents direct independent control of multiple drivers on one I2C bus. The solution was to use a TCA9548A I2C multiplexer. The Arduino selects the required channel before sending commands to the selected haptic driver. For dual-motor feedback, two channels can be enabled at the same time.

### Motor indexing

The position mapping depends on correct motor indexing. During testing, each motor had to be checked individually so that Motor 1, Motor 2, Motor 3, and Motor 4 matched the intended physical locations. Incorrect indexing would make the position feedback confusing.

### Speed feedback clarity

The first feedback approach needed to be clear enough for the user to distinguish safe movement from excessive speed. Pulse feedback was kept below the maximum speed because it gives a graded sensation. Continuous vibration was reserved for speeds above 1 mm/s because it is more noticeable and works well as a warning cue.

### PCB reliability

A soldered PCB was used instead of a breadboard to improve reliability. Since the electronics are mounted on the exoskeleton, a breadboard would be more vulnerable to vibration and movement. The PCB made the system more suitable for repeated tests and physical integration.

### Exoskeleton tuning

The exoskeleton control gains were tuned by trial and error. Too little position gain made the guidance weak, while too much gain risked oscillation or uncomfortable motion. The velocity gain was used to damp the response. The velocity integrator was set to zero because integral action was not required for the training behaviour and could make the system less stable.

## 6. Discussion

The prototype demonstrates a multimodal approach to cochlear implant insertion training by combining a simulated insertion environment, physical motion guidance through an exoskeleton, and vibrotactile feedback through four Titan Haptics Drake motors. Instead of relying only on visual information, the system provides the user with physical and tactile cues that are directly related to the desired insertion behavior. This is relevant for cochlear implant insertion because controlled insertion speed, stable movement, and accurate final positioning are important aspects of atraumatic insertion technique.

A key design decision was to separate position feedback from speed feedback. Position was communicated spatially by activating different motors, or pairs of motors, according to the cyclic position range. Single-motor activation gave a localized cue, while dual-motor activation represented transition regions between two position zones. This allowed four physical motors to represent eight perceptible position sectors. Speed was communicated through the temporal pattern of the vibration. Below the maximum desired insertion speed of 1 mm/s, the motors produced pulses, with the pulse frequency increasing as the speed approached the limit. Above 1 mm/s, the feedback changed to continuous vibration, creating a clear warning that the movement was too fast.

The haptic feedback results suggest that the tactile cues improved the repeatability of the task. The end-position error plot shows a much wider spread in the no-haptic condition, while the haptic condition is more concentrated around the desired final position. This indicates that the position feedback helped the user recognize the intended end region more consistently. The speed standard deviation results also show a narrower distribution with haptic feedback, suggesting that the pulse-based speed cue helped the user maintain a more consistent insertion speed. These results support the idea that vibrotactile feedback can improve user awareness of both position and speed during a simulated insertion task.

The exoskeleton adds another layer of guidance by physically pushing or guiding the user through the desired movement. This is useful for training because the user can feel the target motion instead of only observing it visually. The use of a repeatable Excel/data-sheet trajectory also makes the task more controlled since the same insertion profile can be repeated across trials. From a control perspective, the final ODrive gain values were selected experimentally to balance guidance strength and stability. A position gain of 42.5 provided sufficient movement guidance, a velocity gain of 0.315 added damping, and the velocity integrator was kept at 0 to avoid unnecessary oscillation or unstable behavior.

From an engineering perspective, the project required integration between mechanical design, motor control, embedded electronics, haptic actuation, and simulation. The soldered PCB was an important improvement because the electronics are mounted on a moving exoskeleton, where loose breadboard wiring would be unreliable. The TCA9548A multiplexer was also necessary because the DRV2605L haptic drivers share the same default I2C address. This

allowed the Arduino to control multiple haptic motors independently and to activate either one motor or two motors depending on the position sector.

The main limitation is that the current quantitative testing was performed on the haptic feedback system rather than on the full combined assembly. Therefore, the results show that the feedback strategy is promising, but they do not yet prove that the full exoskeleton-integrated system improves cochlear implant insertion training. Future testing should evaluate the complete system with the Unity/Phantom environment, exoskeleton, PCB, haptic motors, ODrive, motor, and power supply operating together. The current prototype also does not measure real electrode insertion force or intracochlear pressure, so it should be considered a training and demonstration platform rather than a clinically validated device

## 7. Conclusion and future work

This project developed a proof-of-concept haptic feedback system for cochlear implant electrode insertion training. The system combines a Unity/Phantom simulation, an actuated arm exoskeleton, an ODrive-controlled motor, a soldered PCB, and four Titan Haptics Drake motors.

The haptic feedback strategy uses two complementary ideas. First, insertion position is represented using a cyclic four-motor mapping, where motors activate individually or in pairs over 45° sectors. Second, insertion speed is represented using pulse frequency and continuous vibration. Speeds below 1 mm/s produce pulses, with more pulses as the speed approaches the limit. Speeds above 1 mm/s produce continuous vibration as a warning.

The exoskeleton guides the user through the insertion motion, allowing the system to teach the movement physically rather than only visually. The final ODrive tuning values were selected experimentally: position gain 42.5, velocity gain 0.315, and velocity integrator gain 0. This produced a practical guided movement while avoiding unnecessary integral action.

Future improvements could include:

- completing full-system testing with the Unity Phantom environment, exoskeleton, PCB, and haptic motors working together
- adding quantitative user tests comparing performance with and without haptic feedback
- measuring whether users stay below the 1 mm/s speed limit more consistently with feedback
- logging speed, position, active motor state, and exoskeleton position during each trial
- adding plots of exoskeleton target tracking and position error
- improving the mechanical comfort and adjustability of the exoskeleton
- validating the haptic motor position mapping with multiple users
- improving the Unity simulation to better represent cochlear implant insertion constraints



- replacing the Excel/data-sheet input with a real-time closed-loop link from the Phantom environment if needed

Overall, the project demonstrates how haptic technology can support medical skill training by turning important motion variables into direct tactile and physical guidance.

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